

PRASANNA

ROBOTICS ENGINEER

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Robotics Engineer with **2.5+ years of experience** in robotics R&D and hands-on product development. Interested in integrating **robotics hardware and software** to build practical robotic systems, with experience across mechanical design, electronics, embedded control, prototyping, testing, and system integration. Passionate about solving real-world engineering challenges and developing robotic systems from concept to functional prototypes.

TECHNICAL SKILLS

Robotics & ROS 2: ROS 2 Humble · ros2_control · SLAM Toolbox · Nav2 · Gazebo · RViz · URDF/Xacro · TF2 · LiDAR · Differential Drive

Programming: C++ & Python (Actively Learning) · OpenCV · Git · Ubuntu/Linux

Electronics: ESP32 · Arduino · Raspberry Pi · Sensors · Motor Drivers · Servo Control · Embedded Systems · IoT

Mechanical & Fabrication: Fusion 360 · Product Design · FDM 3D Printers · CO₂ Laser Cutting · RDWorks

PROFESSIONAL EXPERIENCE

ZAFI ROBOTS® / PROPELLER TECHNOLOGIES R&D PVT LTD

Robotics Engineer — Research & Development (R&D)

Jun 2025 – Present

- Developed and supported **Zafi 1.0, Zafira, and Zafi 2.0** robotic platforms across mechanical design, fabrication, electronics, assembly, and system integration.
- Contributed to **Zafi 2.0** product development and developed an **Autonomous Mobile Robot (AMR)** across hardware, embedded systems, robotics software, testing, and troubleshooting.
- Performed robot **servicing, maintenance, fabrication improvements, testing, and system-level troubleshooting**.

Design Engineer — Research & Development (R&D)

Dec 2024 – Jun 2025

- Delivered hands-on **3D Printing and Rapid Prototyping training** under the **Naan Mudhalvan program** and conducted **Faculty Development Programs (FDPs)** for engineering professors across Tamil Nadu.
- Designed mechanical components and prototypes for **robotics, IoT, drones, and engineering projects** using CAD and digital fabrication.
- Managed prototyping infrastructure including **3D printers, CO₂ laser systems, and robotics equipment**.

Associate STEM Trainer — Training & Development (T&D)

Jul 2024 – Dec 2024

- Conducted hands-on technology workshops across **20+ districts in Tamil Nadu** covering robotics and emerging technologies.
- Guided students and trainers in **project development, hardware integration, prototyping, and troubleshooting**.

FORGE INNOVATION & VENTURES

Innovation Engineer Trainee

Feb 2024 – Jul 2024

- Developed a **ROS 2-based Autonomous Mobile Robot** as a final-year engineering project covering physical hardware and simulation.
- Worked across **mechanical, electronics, embedded, and software integration** to develop and test the robotic platform.

SELECTED PROJECTS

ZAFI 2.0 — Wheel-Based Humanoid Robot

5.5-ft wheel-based humanoid robot with **5-DOF upper body**. Designed and fabricated mechanical components using Fusion 360, 3D printing, and CO₂ laser cutting; integrated Arduino Mega, ESP32, servo motors, Cytron motor drivers, Raspberry Pi 5, and wireless control.

AUTONOMOUS MOBILE ROBOT — ROS 2

Differential-drive AMR integrating **RPLIDAR A1, ros2_control, URDF/Xacro, TF2, Gazebo, SLAM Toolbox, RViz, and Nav2** for robot modeling, mapping, and autonomous navigation.

AUTONOMOUS SAMPLE COLLECTION ROBOT

Arduino-based autonomous robot using **IR sensors, line following, robotic arm, servo motors, and motor drivers** for automated sample collection.

EDUCATION

B.E.—Electrical & Electronics Engineering | CGPA: **8.67** | Government College of Engineering, Srirangam | 2021–2024

Diploma — Electrical & Electronics Engineering | **94.17%** | Seshasayee Institute of Technology, Trichy | 2018–2020

INTERNSHIPS & TRAINING

Intern — National Small Industries Corporation (NSIC) | Jun 2023 – Jul 2023

- Learned Arduino, sensor interfacing, and developed basic embedded/IoT projects.
- Used Proteus and Tinkercad for circuit design and simulation.